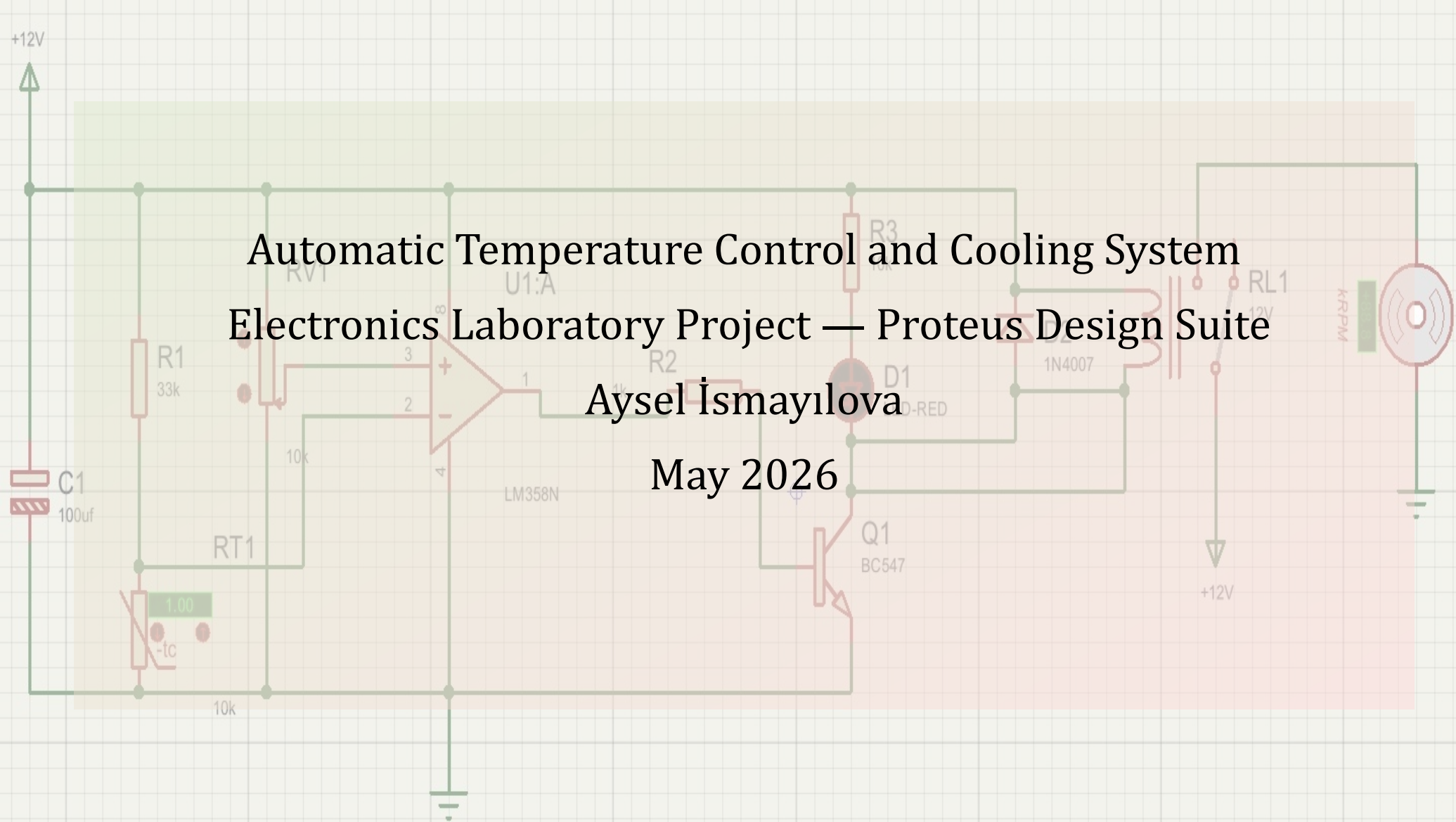




PROJECT OVERVIEW & OBJECTIVE

Design and simulate a circuit that **autonomously activates a cooling fan** when temperature exceeds a pre-set threshold — with **zero human intervention**.



SENSE

NTC Thermistor detects thermal variance



COMPARE

LM358N evaluates voltage thresholds



ACT

Relay triggers DC Motor (Cooling Fan)

REAL-WORLD RELEVANCE

CPU Cooling

HVAC Systems

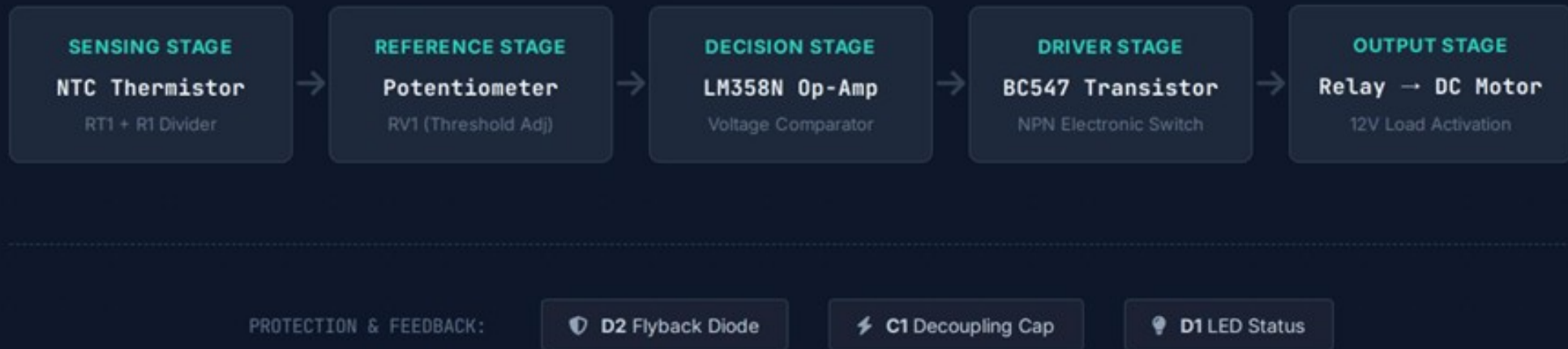
Industrial Thermal Protection

Environment: Proteus Design Suite 8

Served: Understanding of Concepts, Application of Knowledge

SYSTEM ARCHITECTURE

Functional decomposition of the automated cooling loop



COMPONENT LIST & ROLES

COMPONENT	MODEL / VALUE	ROLE IN CIRCUIT
NTC Thermistor	10kΩ	Primary sensor; resistance drops as ambient temperature rises
Potentiometer	RV1 (10kΩ)	Provides an adjustable reference voltage to set the trigger point
R1 Resistor	10kΩ	Forms a voltage divider network with the thermistor
R2 Resistor	1kΩ	Limits base current to the switching transistor
R3 Resistor	1kΩ	Limits forward current to the indicator LED
C1 Capacitor	100nF	Filters out noise and decouples the power supply
★ Op-Amp Comparator	LM358N	Evaluates sensor vs. reference voltage to make binary decisions. <i>Single-supply operation up to 32V makes it highly stable for this 12V unipolar design.</i>
★ NPN Transistor	BC547	Acts as an electronic switch to drive the relay coil. <i>Chosen for $h_{FE} \geq 110$, ensuring sufficient gain to fully saturate the relay current.</i>
LED Indicator	Green 5mm	Provides immediate visual feedback when the fan is active
★ Flyback Diode	1N4007	Protects the transistor from inductive voltage spikes. <i>High reverse voltage tolerance (1000V) safely snubs high-energy inductive kicks.</i>
Electromechanical Relay	12V SPDT	Isolates the low-power logic circuit from the high-power motor
DC Motor	12V Fan Load	The final output device responsible for active cooling

CIRCUIT SCHEMATIC

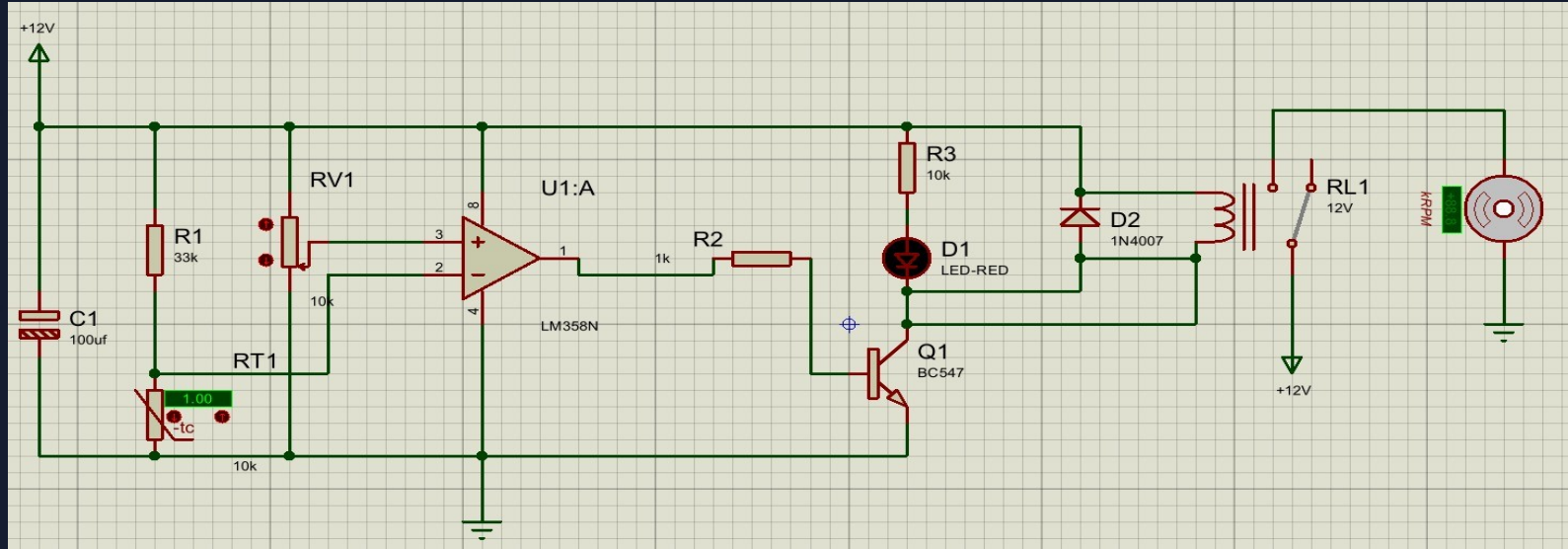
1. SENSING

2. REFERENCE

3. DECISION

4. DRIVER

5. OUTPUT



All stages flow left to right: temperature is sensed, compared, and acted upon automatically.

HOW IT WORKS: THE TWO STATES

STATE 1: COOLING OFF

$T < \text{Threshold}$

SENSING

NTC Resistance: HIGH

$R_{\text{NTC}} \approx 10\text{k}\Omega$ (@ 25°C)

DECISION

$V_- > V_+$

Pin 2 Voltage > Pin 3 (Ref)

COMPARATOR

Output: LOW (0V)

Logic level 0

SWITCHING

Transistor: OFF

Relay contacts open

OUTPUT

Stationary

Motor OFF | LED Dark

STATE 2: COOLING ON

$T > \text{Threshold}$

SENSING

NTC Resistance: LOW

$R_{\text{NTC}} \downarrow$ (e.g. 3kΩ @ 60°C)

DECISION

$V_+ > V_-$

Pin 3 (Ref) > Pin 2 Voltage

COMPARATOR

Output: HIGH (~10V)

Logic level 1

SWITCHING

Transistor: ON

Relay coil energized (contacts close)

OUTPUT

Active Cooling

Motor Running | LED Active

KEY CALCULATIONS

SENSING VOLTAGE AT ROOM TEMPERATURE (V_-)

$$V_- = 12 \times 10,000 / (33,000 + 10,000) \approx 2.79 \text{ V}$$

→ Motor stays OFF

SENSING VOLTAGE AT HIGH TEMPERATURE ($R_{NTC} = 3K\Omega$)

$$V_- = 12 \times 3,000 / (33,000 + 3,000) \approx 1.00 \text{ V}$$

→ Falls below threshold → Motor turns ON

TRANSISTOR BASE CURRENT (CONFIRMS Q1 SAFELY SATURATES RELAY)

$$I_B = (10 - 0.7) / 1,000 = 9.3 \text{ mA}$$

→ Drives $\leq 1.02 \text{ A}$ → Relay needs 50-80 mA ✓

SIMULATION RESULTS

🖥 Environment: Proteus Design Suite 8 (SPICE-based)

CONDITION	OP-AMP PIN 1 OUTPUT	SYSTEM STATE
T < Threshold	~0V (LOW)	Motor OFF, LED OFF
T > Threshold	~10V (HIGH)	Motor ON, LED ON
Relay switching	Flyback spike suppressed	D2 briefly conducts

KEY VOLTAGE OBSERVATIONS AT NODES

Node: Pin 2 (Inverting) 2.8V → Drops below 1.5V	Node: Pin 3 (Non-Inverting) Fixed 1.5V (Reference)	Node: Q1 Collector 12V → ~0V (at saturation)
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PROTECTION FEATURES & DESIGN DECISIONS



Flyback Diode (D2 – 1N4007)

When the relay coil de-energizes, it generates a reverse voltage spike. Without D2, this spike would destroy the BC547 transistor. **D2 provides a safe discharge path.**



Decoupling Capacitor (C1 – 100 μ F)

Motor and relay switching causes voltage dips on the 12V rail. **C1 absorbs these transients**, keeping the Op-Amp reference stable and preventing false triggers.



Design Recommendation: Hysteresis

Adding a $\sim 1\text{M}\Omega$ feedback resistor from Pin 1 to Pin 3 of the LM358N creates a **Schmitt trigger**. This eliminates relay chattering when the temperature hovers exactly at the threshold.

TROUBLESHOOTING & PROBLEM SOLVING

SYMPTOM	ROOT CAUSE	SOLUTION
⚠ Op-Amp output unstable / flickering	No hysteresis – noise at threshold	🔧 Add positive feedback (Schmitt trigger)
💥 Transistor destroyed after relay switch-off	Flyback diode D2 missing or reversed	🔧 Confirm D2 cathode → +12V rail
🔌 Circuit resets when motor starts	Power supply inrush; no C1	🔧 Ensure 100μF capacitor on supply rail

“Anticipating failure modes during design is as important as the design itself.”

REAL-WORLD APPLICATIONS & FUTURE IMPROVEMENTS

REAL-WORLD ADAPTATIONS



Server Room Cooling

Uses the exact same circuit logic. The threshold is adjusted via RV1 for a **~35°C activation point**. Multiple industrial fans could easily be relay-switched in parallel.



Battery Thermal Protection

Applied to EV or drone power packs. The circuit is adapted to **cut the charging relay** if the pack temperature exceeds safe limits, with the LED serving as a safety warning indicator.

SYSTEM EVOLUTION



Proportional PWM Control

Replacing the binary Op-Amp comparator with a PWM (Pulse Width Modulation) controller would drastically increase system efficiency.

Instead of an abrupt ON/OFF state, the motor speed would vary **proportionally with the temperature** — running slowly at mild heat, and at maximum RPM during critical heat events.

NE555 Astable Mode

Microcontroller

Energy Efficient

REFLECTION & CONCLUSION

WHAT I LEARNED

I understood how a single Op-Amp can replace a complex microcontroller for binary control tasks — **simplicity is a design virtue**.

WHAT CHALLENGED ME

Identifying why the relay was chattering — tracing it back to the **absence of hysteresis** was not obvious at first.

WHAT I'D DO DIFFERENTLY

Implement the **Schmitt trigger** from the start, and add a temperature display (e.g., LM35 + voltmeter) to make the threshold visible during calibration.



This project demonstrated a complete closed-loop control system — **SENSE, COMPARE, ACT** — built from fundamental analog components and verified through mathematical calculation and Proteus simulation.